# SKILL 4

## Linear Regression and Regularisation

One Piece Bounty Prediction Project

## Aim

Compare OLS, Ridge, Lasso and ElasticNet regression models using polynomial features, cross-validation and test-set diagnostics.

## Dataset and approach

The target is log10(bounty in Berries). The supplied summary reports 142 rows used; unknown bounty records, two low-bounty outliers and one yen-denominated bounty were excluded as documented. Degree 2 polynomial features were selected using Ridge cross-validation.

## Model comparison

| Model | Train $R^2$ | Test $R^2$ | Test RMSE | CV RMSE | Non-zero |
|---|---|---|---|---|---|
| OLS | 0.962 | 0.857 | 0.254 | 0.302 | 35 |
| Ridge | 0.952 | 0.934 | 0.172 | 0.238 | 35 |
| Lasso | 0.940 | 0.957 | 0.140 | 0.206 | 7 |
| ElasticNet | 0.938 | 0.956 | 0.140 | 0.200 | 7 |

Selected model: ElasticNet, alpha = 0.02219, l1_ratio = 0.99. Test $R^2$ = 0.956; test RMSE = 0.140 in log10 units; typical miss factor = 1.32×.

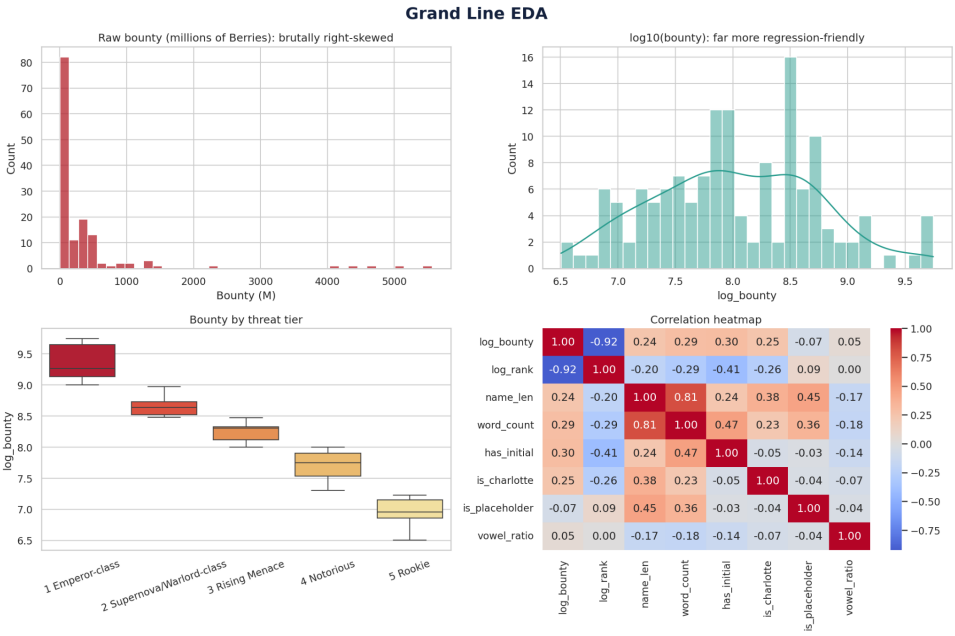
## Key findings

OLS had a larger gap between training and test performance, consistent with overfitting after polynomial expansion. Ridge shrank coefficients while Lasso and ElasticNet set many coefficients exactly to zero. Ablation results were 0.839 ± 0.048 for rank only, 0.073 ± 0.118 for name clues only, and 0.848 ± 0.034 for rank plus name clues.

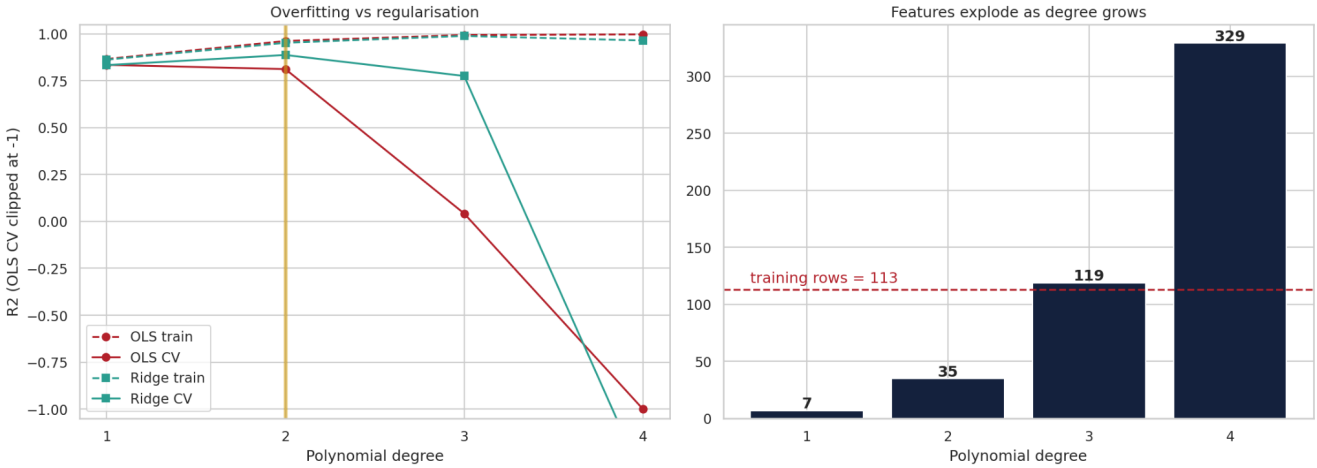
## Limitation

Rank (S.no) is derived from bounty ordering and therefore leaks information about the target. The high score should not be interpreted as proof that name features independently predict bounty well.

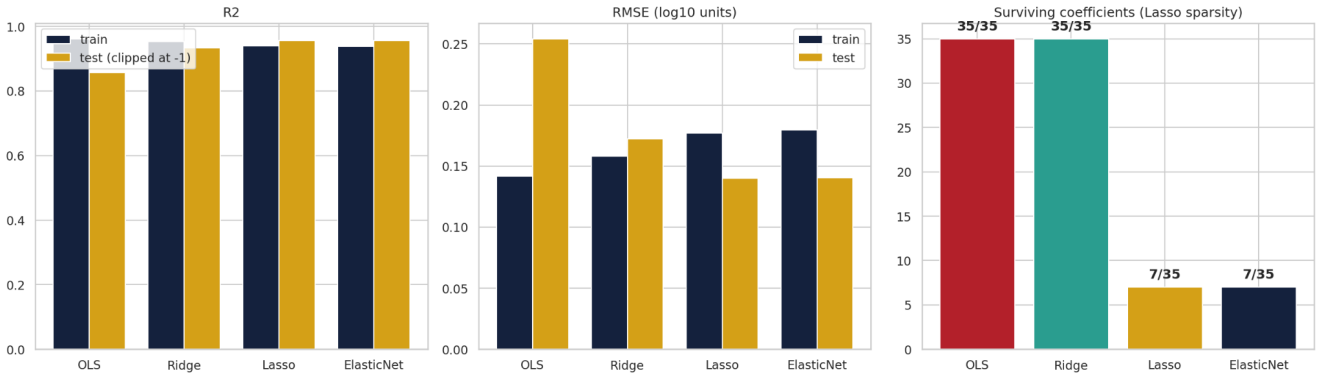
## Exploratory data analysis



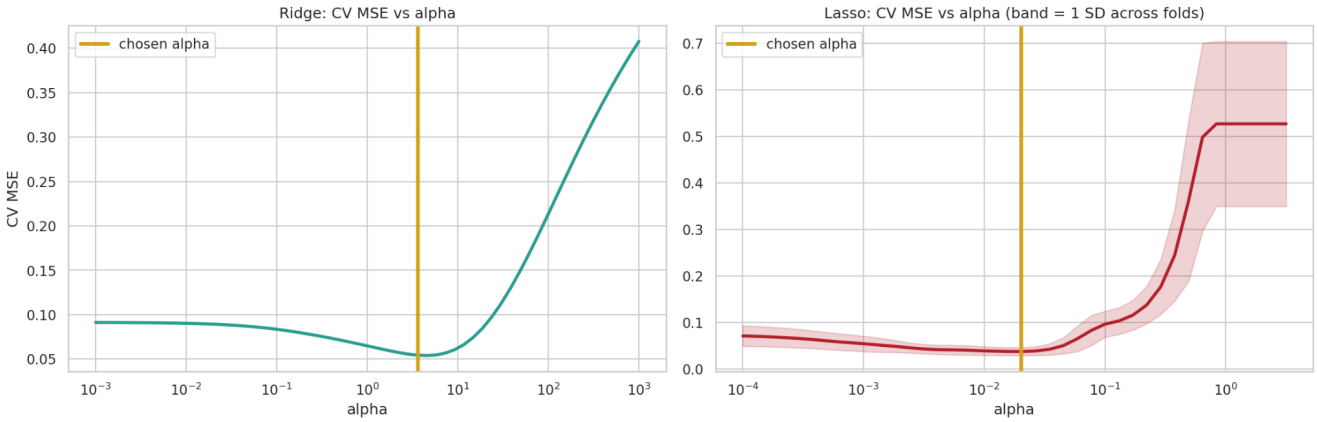

## Polynomial degree and overfitting



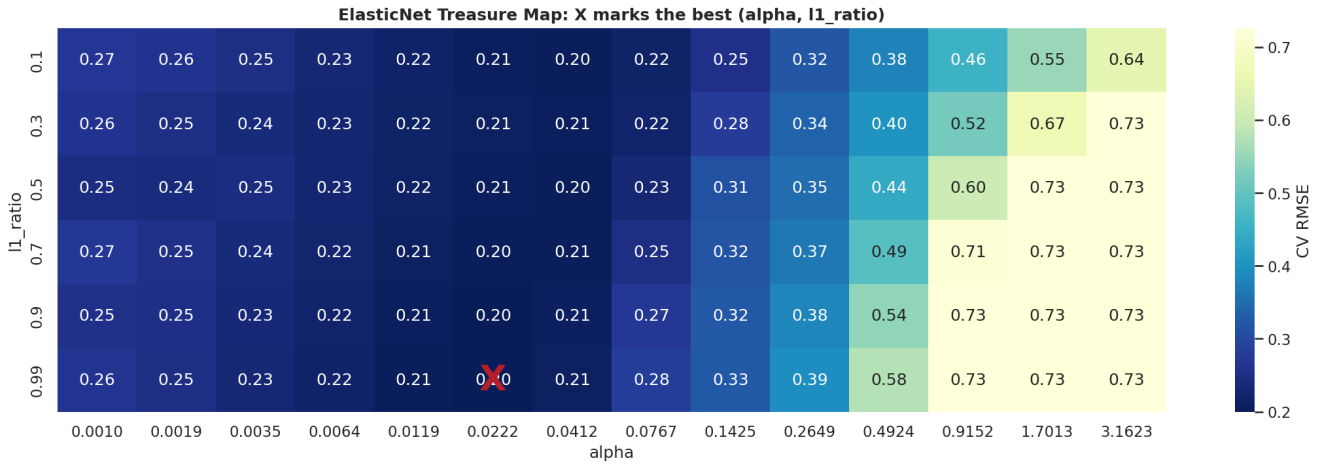

## Model comparison



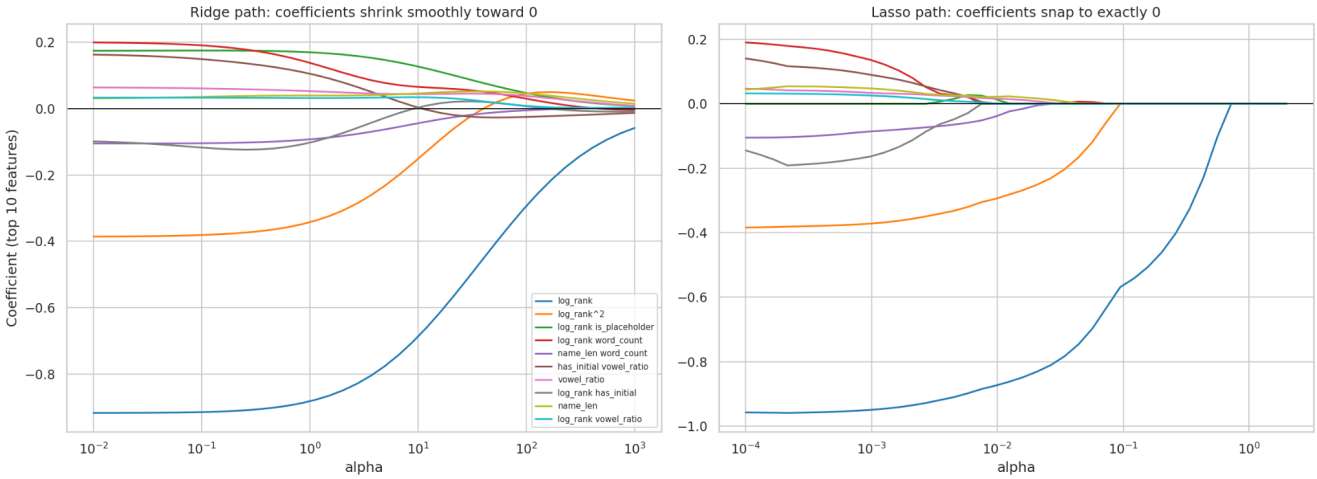

## Regularisation parameter curves



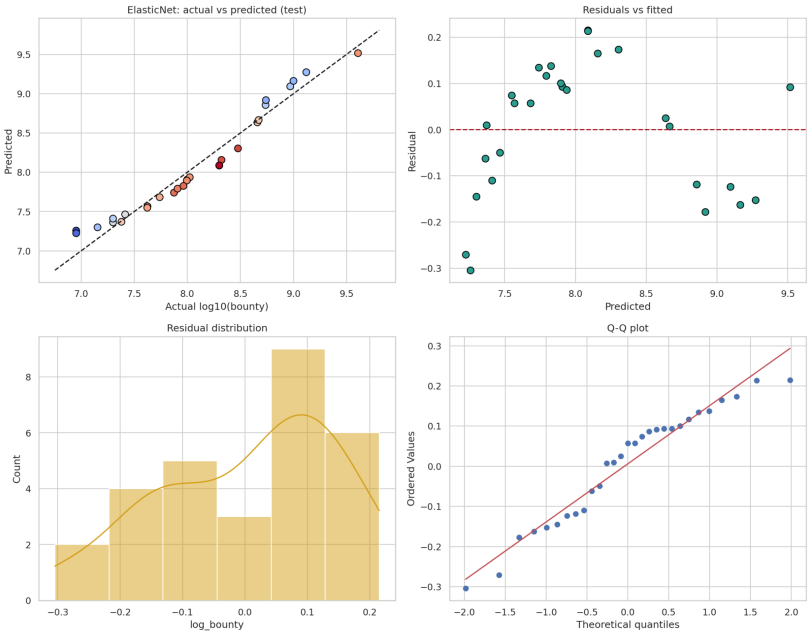

## ElasticNet parameter search




## Coefficient paths




## Prediction and residual diagnostics


ElasticNet: actual vs predicted (test)
Predicted
Actual log10(bounty)
Residuals vs fitted
Residual
Predicted
Residual distribution
Count
log_bounty
Q-Q plot
Ordered Values
Theoretical quantiles